Phase 4

```
        Perfect! Roll 16 means:
```

Your Setting	Value
Your IP	`10.17.100.16`
Your domain	`s16.zhjlab.bd`
Your name server	`ns1.s16.zhjlab.bd`

Phase 4: Making BIND Authoritative for `s16.zhjlab.bd`

We'll create your zone file and configure BIND to answer authoritatively for your domain.

Step 1: Telling BIND about your Zone

First, we have to tell the BIND software: *"Hey, you are now the boss of the domain `s16.zhjlab.bd`."*

Since you are likely on **Ubuntu** for this lab, we use a specific configuration file to declare this.

The Action:

1. Open your terminal.
2. Open the file `/etc/bind/named.conf.local` with `sudo` and a text editor (like `nano` or `gedit`).
3. Add the following block to the bottom of the file:

Bash

```
zone "s16.zhjlab.bd" {
    type master;
    file "/etc/bind/db.s16.zhjlab.bd";
};
```

What this code means:

- `zone "s16.zhjlab.bd"`: This defines the name of your "kingdom."

- `type master;`: This means your server is the **Authoritative** source (the Master). It doesn't ask others for this info.
- `file "/etc/bind/db.s16.zhjlab.bd";`: This tells BIND exactly where the list of records (the "Zone File") is located.

This is a much safer way to do it because starting from a template like `db.local` ensures you keep the correct formatting that BIND expects.

Step 2: Creating the Zone File (The Database)

Why we are doing this:

Instead of typing everything from scratch (where a single missing semicolon can break the whole server), we use a "skeleton" file provided by Ubuntu. We then perform "surgery" on that skeleton to make it represent your specific domain and IP.

1. The Command: Copy the Template

First, we make a copy of the default local loopback template and name it after your domain.

Bash

```
sudo cp /etc/bind/db.local /etc/bind/db.s16.zhjlab.bd
```

2. The Action: Open and Modify

Open your new file:

Bash

```
sudo nano /etc/bind/db.s16.zhjlab.bd
```

3. What to Modify (The "Surgery")

When you open the file, it will look like a generic template. You need to change three specific things:

A. Update the SOA (The Domain Name)

Find the line that says `localhost.` and replace it with your domain name.

- **Change from:** `localhost. root.localhost.`
- **Change to:** `ns1.s16.zhjlab.bd. admin.s16.zhjlab.bd.`

- **Why:** You are telling BIND that `ns1.s16.zhjlab.bd` is now the authority, and `admin@s16.zhjlab.bd` is the contact.

B. Update the Serial Number

Find the number `2` (usually under "Serial").

- **Change to:** `2026021901`
- **Why:** This tells other servers "This is a fresh update from today (Feb 19, 2026)." If you change this file later, increment this to `...02`.

C. Update the Records (The bottom of the file)

Delete the lines at the bottom (the ones with `127.0.0.1` and `::1`) and add your specific server records.

- **Change from:** Plaintext

```
@   IN  NS  localhost.
@   IN  A   127.0.0.1
@   IN  AAAA ::1
```

- **Change to:** Plaintext

```
@       IN      NS      ns1.s16.zhjlab.bd.
ns1     IN      A       10.17.100.16
www     IN      A       10.17.100.16
mail    IN      A       10.17.100.16
```

- **Why:** * The `NS` line tells the world *who* the boss is (`ns1`).
  - The `A` lines tell the world *where* the boss is located (your machine's IP `10.17.100.16`).

The Final Result

Your file should now look exactly like this:

Plaintext

```
$TTL    604800
@       IN      SOA     ns1.s16.zhjlab.bd. admin.s16.zhjlab.bd. (
                        2026021901      ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;
@       IN      NS      ns1.s16.zhjlab.bd.
```

```
ns1     IN      A       10.17.100.16
www     IN      A       10.17.100.16
mail    IN      A       10.17.100.16
```

Step 3: Syntax Verification (The "Spell-Check")

Why we are doing this: BIND is extremely sensitive. If you forgot a semicolon `;`, a bracket `}`, or a tiny dot `.` at the edig @127.0.0.1 www.s16.zhjlab.bdnd of a domain name, the service will simply refuse to start. Instead of restarting the server and having it crash, we use "pre-flight" tools to check our work.

Think of this like a compiler for your code—it finds the bugs before you "run" the program.

The Action:

We need to check two things: the **main configuration** and the **specific zone file**.

1. Check the Main Configuration

Run this command to check `/etc/bind/named.conf.local` (the file from Step 1):

Bash

```
sudo named-checkconf
```

- **What to look for:** If it returns **nothing**, that is perfect. It means your syntax is correct. If it shows an error, it will tell you exactly which line has the mistake.

2. Check the Zone File

Run this command to check the "database" you just created (the file from Step 2):

Bash

```
sudo named-checkzone s16.zhjlab.bd /etc/bind/db.s16.zhjlab.bd
```

- **What to look for:** You want to see: `OK` and `Loaded serial 2026021901`.

Troubleshooting (The "Why" if it fails)

If you get an error, it is usually one of these three things:

1. **Missing Semicolon:** Every line in the configuration (Step 1) must end in a `;`.
2. **Missing Dot:** In the Zone file (Step 2), full domains like `ns1.s16.zhjlab.bd.` **must** end with a dot.
3. **Permissions:** If the tool says "Permission denied," make sure you are using `sudo`.

Step 4: Activating the Authority

Why we are doing this: Even though you've written the files and verified they are correct, BIND is still running with its **old** instructions from Phase 2. To make it "load" your new role as the Chef/Authority, you have to restart the service. This forces BIND to read `/etc/bind/named.conf.local` and your new zone file.

The Action:

1. **Restart BIND:**

   Bash
   ```
   sudo systemctl restart bind9
   ```
2. **Check the status:**

   Bash
   ```
   sudo systemctl status bind9
   ```

What to look for: You want to see `active (running)`. If you see `failed`, it usually means there was a mistake that the check-tools missed (like a file path error).

Step 5: The Final Test (Verification)

Why we are doing this: This is the "moment of truth." We need to prove that your server is actually answering for `s16.zhjlab.bd` and, most importantly, that it is doing so as an **Authoritative** source.

The Action:

Run the `dig` command against your own local server:

Bash

```
dig @127.0.0.1 www.s16.zhjlab.bd
```

How to Read the Output (The "Proof"):

Look closely at the **flags** section of the output.

1. **The Answer:** You should see `www.s16.zhjlab.bd` pointing to `10.17.100.16` in the **ANSWER SECTION**.
2. **The Authority Flag (`aa`):** This is the most important part. In the header, you should see `flags: qr aa rd ra;`.
   - **`aa` stands for "Authoritative Answer."** * If you see `aa`, it means your server is saying: *"I didn't ask anyone else; I own this data."*

What if it doesn't work?

- **No `aa` flag:** You might be hitting a cached result or your zone didn't load. Check `tail -f /var/log/syslog` while restarting BIND to see specific errors.
- **Connection Refused:** BIND might have crashed. Re-run Step 3 checks.

Try the `dig` command now. Do you see the `aa` flag in the output?